WONG CHI FUNG, BRIAN

AI/ML Engineer | LLM Systems | Applied AI R&D

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AI/ML engineer with extensive production R&D experience building and deploying domain-specific AI systems. Skilled in training and fine-tuning large models, developing LLM/RAG applications, model serving, and full-stack delivery.

Experience

AI Research Engineer Oct 2021 – Oct 2025
LifeCode

- Built a sequence-level binding-affinity predictor with LLM encoders, reducing research prediction turnaround from weeks to days.
- Developed toxicity-screening ensembles with XGBoost, CatBoost, Random Forest, and LightGBM to filter high-risk drug candidates before lab work.
- Delivered FastAPI/Docker inference services with logging, metrics, and health monitoring.
- Contributed to the development and deployment of PetaPoly™ 4.0, an AI-driven drug discovery platform supporting internal research workflows.

Frontend Developer Mar 2019 – Feb 2021
theOrigo

- Built responsive production web applications for government, education, healthcare, arts, and enterprise clients.
- Delivered public-facing websites and digital platforms for 1823, EdUHK, Hong Kong Philharmonic, CUHK Medical Clinic, and CUHK Medical Centre.
- Worked with designers, backend engineers, and stakeholders; resolved production issues and supported reliable releases.

Research Assistant Aug 2017 – Jun 2018
The Chinese University of Hong Kong

- Conducted plant-genomics research using DNA/RNA extraction, PCR, and ChIP-seq under rigorous laboratory protocols.

Open-source Projects

Multimodal Chatbot (github.com/brian546/QA_bot)
FastAPI • Streamlit • LangGraph • RAG • Ollama
Grounded multi-document QA supporting PDF, TXT, PPTX, XLSX, and images. Uses BM25 plus semantic retrieval with reciprocal-rank fusion, cited answers, chat history, and session management.

Cross-Lingual LawRAG (github.com/brian546/lawRAG)
LangGraph • RAG • Transformers • MLflow
English-to-German legal RAG agent with multilingual bi-encoder retrieval, cross-encoder reranking, exact-citation prediction, and Macro-F1, Precision@k, and Recall@k evaluation.

Education

MSc, Information Technology (Distinction)
Hong Kong Polytechnic University 2024–2026

MSc, Genomics & Bioinformatics
Chinese University of Hong Kong 2016–2017

BSc, Biochemistry & Cell Biology
Hong Kong University of Science and Technology 2012–2016

Training

Data Science & Machine Learning Immersive
Xccelerate Mar 2021 – Jun 2021

Software Engineering Immersive
Xccelerate Jul 2018 – Feb 2019

Core Skills

Programming Languages
Python, JavaScript, SQL, C++

LLM & Retrieval
OpenAI, LangChain, LangGraph, RAG, FAISS, pgvector, BM25, hybrid retrieval, evaluation

ML & Data
PyTorch, Transformers, scikit-learn, Pandas, NumPy, SciPy, XGBoost, CatBoost, LightGBM, MLflow, LoRA, QLoRA

Engineering
FastAPI, Docker, React, Node.js, Streamlit, REST APIs, Git, GitHub